

Linear constraints

Standard form

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Standard form

*It is convenient to write linear constraints in standard form.
All inequality constraints are non negativity constraints.
The rest are equality constraints.*

Standard form

$$\min_{x \in \mathbb{R}^n} f(x)$$

subject to

$$Ax = b$$

$$x \geq 0$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

Equality constraints

$$Ax = b$$

Number of x such that $Ax = b$

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- ▶ 0: incompatible
- ▶ 1: non singular
- ▶ ∞ : underdetermined *The only interesting one*

Rank

$$A = \begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & -1 & 1 & 0 \end{pmatrix} \quad b = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} \quad \text{rank}(A) = 2.$$

Compatible: $x_1 = 0, x_2 = 0, x_3 = 5, x_4 = 2.$

$$x_1 - x_2 + x_4 = 2$$

$$x_3 - x_4 = 3$$

$$x_1 - x_2 + x_3 = 5$$

$$x_4 = 2 - x_1 + x_2$$

$$x_3 - 2 + x_1 - x_2 = 3$$

Redundant constraints

- ▶ Consider a compatible system $Ax = b$, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$.
- ▶ $\text{Rank}(A) = r < m$.
- ▶ Then there exists $m - r$ redundant constraints that can be removed.

Theorem 3.6

It can be assumed that A is of full rank.

Summary

It is convenient to write the constraints in standard form

The system of linear equations can be assumed to be compatible (and underdetermined) and of full rank.

The remaining constraints are non negativity constraints.

Identifying active constraints amount to find the variables that are zero at the solution.